

UNIT-3

Wireless characteristics

Overview

Wireless communication is both a blessing and a curse for sensor networks. On the one hand, it is key to their flexible and low-cost deployment. On the other hand, it imposes considerable challenges because wireless communication is expensive and wireless link conditions are often harsh and vary considerably in both space and time due to multi-path propagation effects. We will focus on three sets of simple models that are useful in understanding and analyzing higher-layer networking protocols for WSN:

1. **Link quality model:** a realistic model showing how packet reception rate varies statistically with distance. This incorporates both an RF propagation model and a radio reception model.
2. **Energy model:** a realistic model for energy costs of radio transmissions, receptions, and idle listening.
3. **Interference model:** a realistic model that incorporates the capture effect whereby packets from high-power transmitters can be successfully received even in the presence of simultaneous traffic.

Wireless link quality

The following is the basic ideal model of a wireless link: two nodes have a perfect link (with 100% packet reception rate) if they are within communication range R , and a non-existent link (0% packet reception rate) if they are outside this range. This ideal model is the basis of many algorithms and is widely used in analytical and simulation studies. While it is useful in some contexts, the ideal model can be quite misleading when designing and evaluating routing protocols. Therefore we seek more realistic models based on real-world observations.

Empirical observations

Several researchers have undertaken experimental studies of link quality in WSN. The following are several key observations from these studies:

1. The packet reception contour formed by receptions at different locations from the same transmitter is not regular or isotropic (see Figure 5.1).

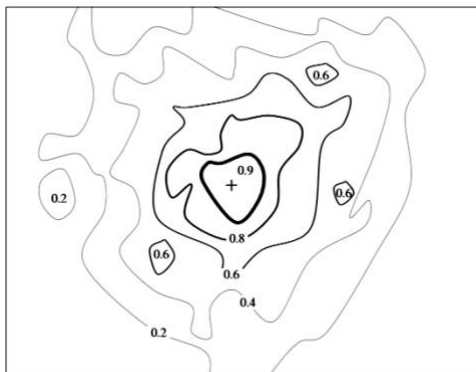


Figure 5.1 A realistic packet reception rate contour

2. The link quality distributions with and without power control are highly dependent on environment and individual hardware differences. Indoor office environments, for instance, show worse link quality

distributions than clutter free outdoor settings. Changing the transmitter or receiver at the same location can change the link quality.

3. There are three distinct regions of link quality: a nearby connected region where packet reception rates are consistently high; beyond this lies a transitional region (also referred to as a gray area), where reception rates are highly variant (see Figure 5.2).

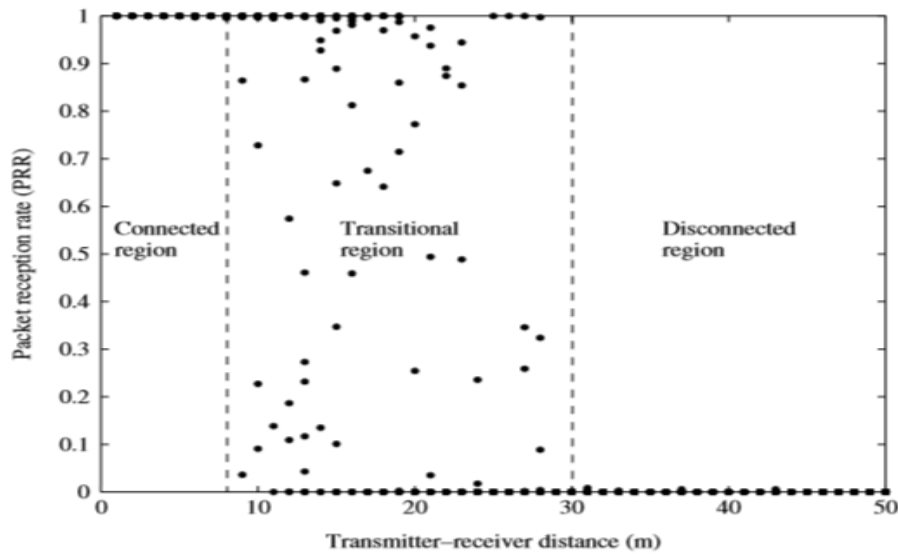


Figure 5.2 Realistic packet reception rate statistics with respect to inter-node distance

4. If a link has a very high packet reception rate (>99%), then the link is firmly within the connected region and is likely to be quite reliable over time

Radio energy considerations

Table 5.2 shows typical power consumption costs for a number of radio hardware devices for the different operating modes, as well as the time taken for turning the radio on to an active mode from sleep mode. The time to switch a radio back to sleep mode is not shown in this table, but it is generally negligible (on the order of 10 s). While various radios differ in the absolute power values, depending on the output power and range, a common feature that should be observed in their power costs is that sleep mode generally consumes three orders of magnitude less power, while receive power and transmit power costs are of the same magnitude, with the receive-transmit power ratio ranging from about 1:1 to 1:2. The power costs for (a) keeping the mode in receive mode without actively receiving packets, (b) keeping the node in receive mode, while receiving packets, and for (c) keeping the node in receive mode while overhearing packets intended for others, are often very similar (within about 20–30%). These observations suggest that it is best to keep a radio in the sleep mode as much as possible (to cut down on idle receive mode costs) and for a long duration each time (to minimize switching costs).

Table 5.2 Typical power consumption costs and startup times for different radios

Radio	Frequency/Data Rate	Sleep	Receive	Transmit	Startup
CC 2420 [18]	2.4 GHz, 250 kbps	60 μ W	59 mW	52 mW	0.6 ms
CC 1000 [17]	868 MHz, 19.2 kbps	0.6 μ W	29 mW	50 mW	2.0 ms
MIT μ AMPS-1 [139]	2.4 GHz, 1 Mbps	negligible	279 mW	330 mW	0.5 ms
IEEE 802.11b [165]	2.4 GHz, 11 Mbps	negligible	1.4 W	2.25 W	1.0 ms

The SINR capture model for interference

In this model, the interference caused by a node's transmissions is modeled to be the same as its ideal circular communication range (see Figure 5.6(a)). No other node within this range may receive a packet successfully due to collisions, while this node is transmitting. A variant of this model is one in which the disc of interference has a radius that is larger than for the disc of communication. While these models – in which good communication on a link implies that the transmissions on that link will always collide with other traffic incoming to the receiving node – offer very simple abstractions that are useful for design and analysis, they can also be potentially highly misleading. In practice, radios are capable of receiving a packet error free, even when another packet is being transmitted by a neighbor (see Figure 5.6(b)). This is known as the capture effect. The following is a realistic model incorporating the capture effect.

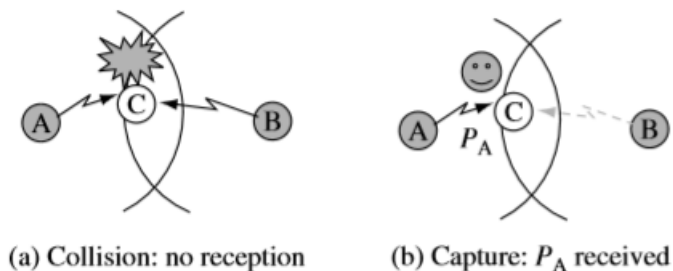


Figure 5.6 Interference in wireless medium: (a) an idealized model, (b) the capture effect model. In the capture effect model simultaneous successful receptions are possible so long as SINR is sufficiently high at each receiver

Medium-access and sleep scheduling

Overview

An essential characteristic of wireless communication is that it provides an inherently shared medium. All medium-access control (MAC) protocols for wireless networks manage the usage of the radio interface to ensure efficient utilization of the shared bandwidth. MAC protocols designed for wireless sensor networks have an additional goal of managing radio activity to conserve energy. Thus, while traditional MAC protocols must balance throughput, delay, and fairness concerns, WSN MAC protocols place an emphasis on energy efficiency as well.

Traditional MAC protocols

We begin with a focus on contention-based MAC protocols. Contention-based MAC protocols have an advantage over contention-free scheduled MAC protocols in low data rate scenarios, where they offer lower latency characteristics and better adaptation to rapid traffic variations.

1. Aloha and CSMA

The simplest forms of medium-access are unslotted Aloha and slotted Aloha. In unslotted Aloha, each node behaves independently and simply transmits a packet whenever it arrives; if a collision occurs, the packet is retransmitted after a random waiting period. The slotted version of Aloha works in a similar manner, but allows transmissions only in specified synchronized slots. Another classic MAC protocol is the carrier sense medium-access (CSMA) protocol. In CSMA, a node that wishes to transmit first listens to the channel to assess whether it is clear. If the channel is idle, the node proceeds to transmit. If the channel is busy, the node waits a random back-off period and tries again. CSMA with collision detection is the basic technique used in IEEE 802.3/Ethernet.

2. Hidden and exposed node problems

Traditional CSMA fails to avoid collisions and is inefficient in wireless networks because of two unique problems: the hidden node problem and the exposed node problems. The hidden node problem is illustrated in Figure 6.1(a); here, node A is transmitting to node B. Node C, which is out of the radio range of A, will sense the channel to be idle and start packet transmission to node B too. In this case CSMA fails to avoid the collision because A and C are hidden to each other. The exposed node problem is illustrated in Figure 6.1(b). Here, while node B is transmitting to node A, node C has a packet intended for node D. Because node C is in range of B, it senses the channel to be busy and is not able to send. However, in theory, because D is outside of the range of B, and A is outside of the range of C, these two transmissions would not collide with each other. The deferred transmission by C causes bandwidth wastage.

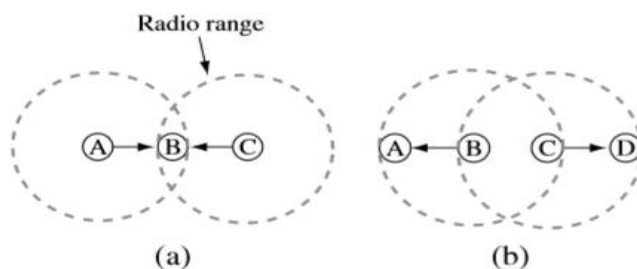


Figure 6.1 Problems with basic CSMA in wireless environments: (a) hidden node, (b) exposed node

These problems are duals of each other in a sense: in the hidden node problem packets collide because sending nodes do not know of another ongoing transmission, whereas in the exposed node problem there is a wasted opportunity to send a packet because of misleading knowledge of a non-interfering transmission. The key underlying mismatch is that it is not the transmitter that needs to sense the carrier, but the receiver. Some communication between the transmitter and receiver is needed to solve these problems.

3. Medium-access with collision avoidance (MACA)

The MACA Protocol by introduced the use of two control messages that can (in principle) solve the hidden and exposed node problems. The control messages are called request to send (RTS) and clear to send (CTS). The essence of the scheme is that when a node wishes to send a message, it issues an RTS packet to its intended recipient. If the recipient is able to receive the packet, it issues a CTS packet. When the sender receives the CTS, it begins to transmit the packet. When a nearby node hears an RTS addressed to another node, it inhibits its own transmission for a while, waiting for a CTS response. If a CTS is not heard, the node can begin its data transmission. If a CTS is received, regardless of whether or not an RTS is heard before, a node inhibits its own transmission for a sufficient time to allow the corresponding data communication to complete. Under a number of idealized assumptions (e.g., ignoring the possibility of RTS/CTS collisions, assuming bidirectional communication, no packet losses, no capture effect) it can be seen that the MACA scheme can solve both the hidden node problem and the exposed node problem.

Using the earlier examples, it solves the hidden node problem because node C would have heard the CTS message and suppressed its colliding transmission. Similarly it solves the exposed node problem because, although node C hears node B's RTS, it would not receive the CTS from node A and thus can transmit its packet after a sufficient wait.

4. IEEE 802.11 MAC

Closely related to MACA is the widely used IEEE 802.11 MAC standard [95]. The 802.11 device can be operated in infrastructure mode (single-hop connection to access points) or in ad hoc mode (multi-hop network). It also includes two mechanisms known as the distributed coordination function (DCF) and the point coordination function (PCF). The DCF is a CSMA-CA protocol (carrier sense multiple access with collision avoidance) with ACKs. A sender first checks to see if it should suppress transmission and back off because the medium is busy; if the medium is not busy, it waits a period DIFS (distributed inter-frame spacing) before transmitting.

The receiver of the message sends an ACK upon successful reception after a period SIFS (short inter-frame spacing).

The RTS/CTS virtual carrier sensing mechanism from MACA is employed, but only for unicast packets. Nodes which overhear RTS/CTS messages record the duration of the entire corresponding DATA-ACK exchange in their NAV (network allocation vector) and defer access during this duration. An exponential backoff is used

- (a) when the medium is sensed busy,
- (b) after each retransmission (in case an ACK is not received), and
- (c) after a successful transmission. In the second mechanism, PCF, a central access point coordinates medium access by polling the other nodes for data periodically. It is particularly useful for real-time applications because it can be used to guarantee worst-case delay bounds.

5. IEEE 802.15.4 MAC

The IEEE 802.15.4 standard is designed for use in low-rate wireless personal area networks (LR-WPAN), including embedded sensing applications. Most of its unique features are for a beacon-enabled mode in a star topology. In the beacon-enabled mode for the star topology, the IEEE 802.15.4 MAC uses a superframe structure shown in Figure 6.2. A superframe is defined by a periodic beacon signal sent by the PAN coordinator. Within the superframe there is an active phase for communication between nodes and the PAN coordinator and an inactive phase, which can be adjusted depending on the sleep duty cycle desired. The active period has 16 slots that consist of three parts: the beacon, a contention access period (CAP), and a collision-free period (CFP) that allows for the allocation of guaranteed time slots (GTS). The presence of the collision-free period allows for reservation-based scheduled access. Nodes which communicate only on guaranteed time slots can remain asleep and need only wake-up just before their assigned GTS slots. The communication during CAP is a simple CSMA-CA algorithm, which allows for a small backoff period to reduce idle listening energy consumption. A performance evaluation of this protocol and its various settings and parameters can be found in [1]. While the IEEE 802.15.4 can, in theory, be used for other topologies, the beacon-enabled mode is not defined for them. In the rest of the chapter we will concern ourselves with both contention-based and schedule-based energy efficient MAC protocols that are relevant to multi-hop wireless networks.

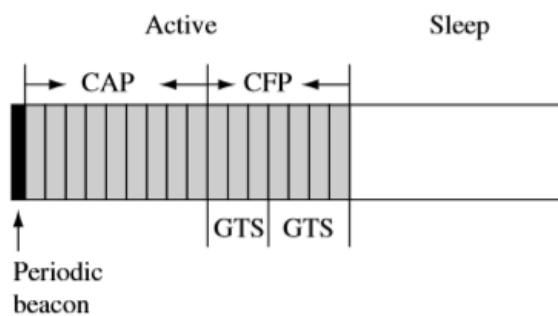


Figure 6.2 The superframe structure of IEEE 802.15.4 MAC

Energy efficiency in MAC protocols

Energy efficiency is obtained in MAC protocols essentially by turning off the radio to sleep mode whenever possible, to save on radio power consumption.

1. Power management in IEEE 802.11

There exist power management options in the infrastructure mode for 802.11. Nodes inform the access point (AP) when they wish to enter sleep mode so that any messages for them can be buffered at the AP. The nodes periodically wake-up to check for these buffered messages. Energy savings are thus provided at the expense of lower throughput and higher latency.

2 Power aware medium-access with signalling (PAMAS)

The PAMAS (power aware multi-access protocol with signalling) [199] is an extension of the MACA technique, where the RTS/CTS signalling is carried out on a separate radio channel from the data exchange. It is one of the first power aware MAC protocols proposed for multi-hop wireless networks. In PAMAS, nodes turn off the radio (go to sleep) whenever they can neither receive nor transmit successfully. Specifically they go to sleep whenever they overhear a neighbor transmitting to another node, or if they determine through the control channel RTS/CTS signaling that one of their neighbors is receiving. The duration of the sleep mode is set to the length of the ongoing transmissions indicated by the control signals received on the secondary channel. If a transmission is started while a node is in sleep mode, upon wake-up the node sends probe signals to determine the duration of the ongoing transmission and how long it can go back to sleep. In PAMAS, a node will only be put to sleep when it is inhibited from transmitting/receiving anyway, so that the delay/throughput performances of the network are not affected adversely. However, there can still be considerable energy wastage in the idle reception mode (i.e. the condition when a node has no packets to send and there is no activity on the channel).

3. Minimizing the idle reception energy costs

While PAMAS provides ways to save energy on overhearing, further energy savings are possible by reducing idle receptions. The key challenge is to allow receivers to sleep a majority of the time, while still ensuring that a node is awake and receiving when a packet intended for it is being transmitted. Based on the methods to solve this problem, there are essentially two classes of contention based sensor network MAC protocols. The first approach is completely asynchronous and relies solely on the use of an additional radio or periodic low-power listening techniques to ensure that the receiver is woken up for an incoming transmission intended for it. The second approach, with many variants, uses periodic duty-cycled sleep schedules for nodes. Most often the schedules are coordinated in such a way that transmitters know in advance when their intended receiver will be awake.

Asynchronous sleep techniques

In these techniques nodes normally keep their radios in sleep mode as a default, waking up briefly only to check for traffic or to send/receive messages.

1. Secondary wake-up radio

Nodes need to be able to sleep to save energy when they do not have any communication activity and be awake to participate in any necessary communications. The first solution is a hardware one – equipping each sensor node with two radios. In such a hardware design, the primary radio is the main data radio, which remains asleep by default. The secondary radio is a low-power wake-up radio that remains on at all times. Such an idea is described in the Pico Radio project [166], as well as by Shih et al. [195]. If the wake-up radio of a node receives a wake-up signal from another node, it responds by waking up the primary radio to begin receiving. This ensures that the primary radio is active only when the node has data to send or receive. The underlying assumption motivating such a design is that, since the wake-up radio need not do much

sophisticated signal processing, it can be designed to be extremely low power. A tradeoff, however, is that all nodes in the broadcast domain of the transmitting node may be woken up.

2. Low-power listening/preamble sampling

El Hoiydi [83] and Hill and Culler [82] independently developed a similar rendezvous mechanism for waking up sleeping radios. In this technique, referred to as preamble sampling or low-power listening, the receivers periodically wake-up to sense the channel. If no activity is found, they go back to sleep. If a node wishes to transmit, it sends a preamble signal prior to packet transmission.

Upon detecting such a preamble, the receiving node will change to a fully active receive mode. The technique is illustrated in Figure 6.3. The wake-up signal could potentially be sent over a high-level packet interface; however, a more efficient approach is to implement this directly in the physical layer – thus the wake-up signal may be no more than a long RF pulse. The detecting node then only checks for the radio energy on the channel to determine whether the signal is present. Hill and Culler argue that this can reduce the receiver check duty cycle to as low as 0.00125%, allowing an almost 2000-fold improvement in lifetime compared with a packet-level wake-up (from about a week to about 38 years). We should note that this scheme will also potentially wake-up all possible receivers in a given transmitter's neighborhood, though mechanisms such as information in the header can be used to put them back to sleep if the communication is not intended for them.

3. WiseMAC

Low-power listening/preamble sampling has one potential shortcoming: the long preamble that the transmitter needs to send can in some situations cause throughput reduction and energy wastage for both sender and receiver. The WiseMAC protocol builds upon preamble sampling to correct this deficiency. Using additional contents of ACK packets, each node learns the periodic sampling times of its neighboring nodes, and uses this information to send a shorter wake-up preamble at just the right time. The preamble duration is determined by the potential clock drift since last synchronization. Let T_W be the receiver sampling period, the clock drift, and L the interval between communications, then the duration of the preamble T_P need only be:

$$T_P = \min(4L\theta, T_W)$$

The packets in WiseMAC also contain a “more” bit (this is also found in the IEEE 802.11 power save protocol), which the transmitter uses to signal to the receiver if it needs to stay awake a little longer in order to receive additional packets intended for it.

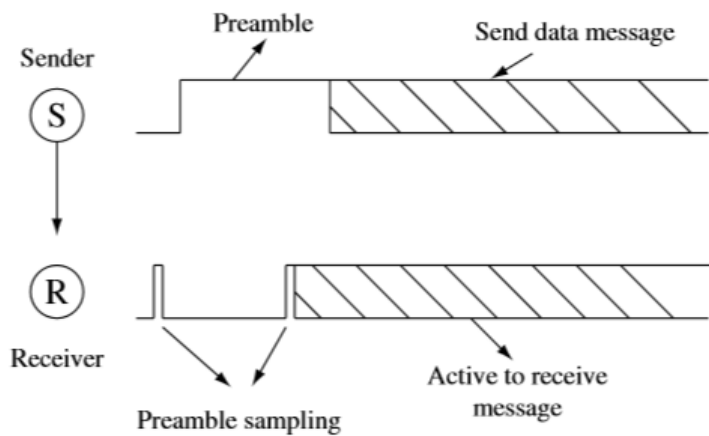


Figure 6.3 The low-power listening technique of preamble sampling

4. Transmitter/receiver-initiated cycle receptions (TICER/RICER)

Similar in spirit to lower-power listening/preamble sampling are the TICER/ RICER techniques [120]. In the transmitter-initiated cycle receiver technique (TICER) (see Figure 6.4(a)), as in low-power listening, the receiver node wakes up periodically to monitor the channel for signals from the sender (which is a wake-up request to send (RTS) signal). The sender sends a sequence of such

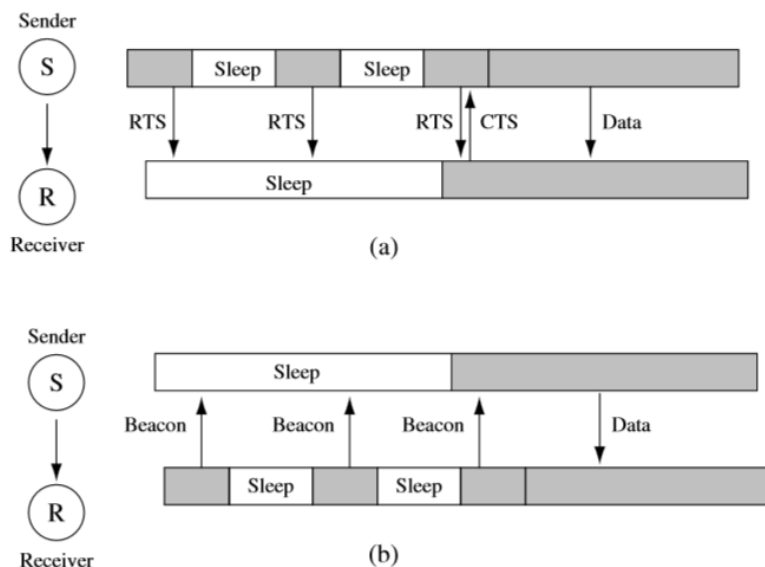


Figure 6.4 Asynchronous sleep using (a) TICER and (b) RICER

RTS signals followed by a short time when it monitors the channel. When the receiver detects an RTS, it responds right away with a CTS signal. If the sender detects a CTS signal in response to its RTS, it begins transmission of the packet. Thus the key difference from preamble sampling is that in TICER the sender sends a sequence of interrupted signals instead of a single long preamble, and waits for an explicit signal from the receiver before transmitting. In the receiver-initiated cycle receiver technique (RICER), illustrated in Figure 6.4(b), a receiving node periodically wakes up to execute a three phase monitor–send wake-up beacon–monitor sequence. A source that wishes to transmit wakes up and stays in a monitoring state. When it hears a wake-up beacon from a receiver, it begins transmission of the data. The receiver in a monitor state that sees the

start of a data packet remains on until the packet reception is completed. However, one subtlety pertaining to the TICER/RICER techniques is that, while the RTS/CTS/wake-up signals involved are easy to implement at a higher packet level, it can be more challenging to do so at a lower-power RF analog level. This is because to match the transmission to the correct receiver, the receiver needs to uniquely identify itself to the transmitter.

5. Reconfigurable MAC protocol (B-MAC)

A nice implementation of this class of low-power MAC protocols is the highly reconfigurable and light-weight B-MAC protocol [161]. It provides a limited set of core functionality and an interface that allows the core components to be tuned and configured depending on higher-layer needs. The core of B-MAC consists of the following features which can be turned on/off and used in any combination desired:

- Low-power listening (LPL), which implements the preamble-based wake-up technique described above, to permit nodes to have sleep as the default mode, helping to conserve energy. Different channel sampling durations and preamble durations can be selected by the higher layers.
- Clear channel assessment (CCA), which determines whether the channel is busy or not by examining multiple adjacent samples and using an appropriate outlier detection technique. If CCA is disabled, a scheduling protocol may be implemented above B-MAC. If it is enabled, the back off duration (in case a busy channel is detected) may be selected by the higher layer. CCA is used for low-power listening.
- Acknowledgements (ACK): If acknowledgements are enabled, a response is sent immediately after receiving any unicast packet.

B-MAC Components		ROM	RAM
Basic B-MAC		3046	166
Basic B-MAC	LPL	4092	170
Basic B-MAC	LPL ACK	4386	172
Basic B-MAC	LPL ACK RTS/CTS	4616	277

Figure 6.5 Components of B-MAC and their memory requirements (in Bytes)

Channel reservation signals for RTS/CTS can be implemented above B-MAC using its control interfaces, but are not part of the core B-MAC itself. The simplicity, efficiency, and configurability of B-MAC (see Figure 6.5) mean that it is likely to be useful in many practical settings. A desirable feature of B-MAC is that it is designed in a modular fashion and provides core functionality. This enables easy implementation of other more complex MAC protocols over B-MAC, including many of the sleep-cycle-based and contention-free time-scheduled protocols that we describe below.

Sleep-scheduled techniques

1. Sensor MAC (S-MAC)

The S-MAC protocol is a wireless MAC protocol designed specifically for WSN. As shown in Figure 6.6, it employs a periodic cycle, where each node sleeps a while, and then wakes up to listen for an interval. The duty cycle of this listen–sleep schedule, which is assumed to be the same for all nodes, provides for a guaranteed reduction in energy consumption. During initialization, nodes remain awake and wait a random period to listen for a message providing the sleep–listen schedule of one of their neighbors. If they do not receive such

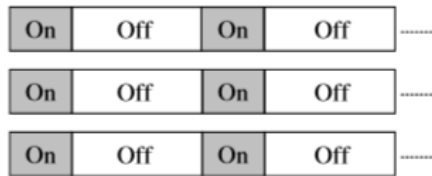


Figure 6.6 Sleep–wake duty cycles in S-MAC

a message, they become synchronizer nodes, picking their own schedules and broadcasting them to their neighbors. Nodes that hear a neighbor’s schedule adopt that schedule and are called follower nodes.

2. Timeout MAC (T-MAC)

The timeout MAC (T-MAC) is a duty-cycled protocol similar in many respects to S-MAC, and like adaptive listening allows modification of the duty cycle. The length of each cycle is kept constant, but the end of the active period is determined dynamically by the use of a timeout mechanism. If a receiver does not receive any messages (data or control) during the timeout interval, it goes to sleep; if it receives such a message, the timer starts afresh after the reception of the message. This renewal mechanism allows for easy adaptation to spatio-temporal variations in traffic. The basic T-MAC scheme suffers from the so-called early sleep problem, which can reduce throughput, particularly in the case of unidirectional flows. When a node has to be silent due to contention in a given cycle, it is unable to send any message to its intended receiver to interrupt its timeout. When the sender can send after the end of the contention period, the intended receiver is already in sleep mode. Two possible solutions to the early sleep problem are proposed and studied in ; we mention them only briefly here. The first solution uses an explicit short FRTS (future request to send) control message that can be communicated to the intended recipient asking it to wait for an additional timeout period. The second solution is called “full buffer priority,” in which a node prefers sending to receiving when its buffer is almost full. With this scheme, a node has higher priority to send its own packet instead of receiving another packet, and is able to interrupt the timeout of its intended receiver.

3. Data-gathering MAC (D-MAC)

For packets that need to traverse multiple hops, both S-MAC and T-MAC provide energy savings at the expense of increased delay. This is because the packet can traverse only a few hops in each cycle before it reaches a node that must go to sleep. This is referred to as the data-forwarding interruption problem. An application-specific solution to this problem is provided by the D-MAC

(data-gathering MAC) protocol, which applies only to flows on a predetermined data-gathering tree going up from the various network nodes to a common sink. D-MAC essentially applies a staggered sleep schedule, where nodes at each successive level up the tree follow a receive–transmit–sleep sequence that is shifted to the right. These cycles are aligned so that a node at level k is in the receiving mode when the node below it on the tree at level $k+1$ is transmitting. This is illustrated in Figure 6.7.

The staggered schedule of D-MAC has many advantages – it allows data and control packets (such as requests for adaptive extensions of the active period) to sequentially traverse all the way up a tree with minimum delay; it allows

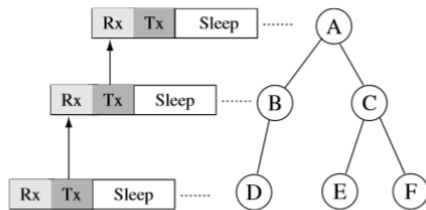


Figure 6.7 The staggered sleep schedule in D-MAC

requests for adaptive extensions of the active period to be propagated all the way up the tree; it reduces interference by separating active periods at the different levels; and it is also shown to reduce the number of nodes that need to be awake when cycle adaption occurs. To deal with contention and interference, D-MAC also includes optional components referred to as data prediction and the use of more-to-send (MTS) packets. Despite these advantages, D-MAC in itself is not a general purpose MAC as it applies only to one-way data-gathering trees. However, as we discuss next, the notion of scheduling the wake-up times in such a way as to minimize delay can be extended to other settings.

4. Delay-efficient sleep scheduling (DESS)

The problem of delay-efficient scheduling of sleep cycles is formulated as a combinatorial optimization problem by Lu et al. The formulation is as follows: each node picks a unique slot out of k slots to use as a reception slot and publishes this to its neighbors. Now, should any of them wish to transmit to this node in any cycle, they wake-up at this slot during the cycle to transmit. On a multi-hop path, the delay that a packet will encounter at each hop is then purely a function of the reception slot times of the corresponding two nodes. For a packet to be sent from node i to its adjacent node j , which have reception slots x_i and x_j respectively, the delay is given by $x_j - x_i \bmod k$ if $x_i \neq x_j$, k otherwise. The path delay is the sum of all the per-hop delays along a given path. The end-to-end delay between the two nodes is the minimum path delay.

Contention-free protocols

Contention-free scheduled medium-access protocols for multi-hop wireless networks date back to the early studies on packet radio networks. These consist of the allocation of time slots for time division multiple-access (TDMA) techniques (or frequency channels for FDMA, distinct codes/hopping sequences for CDMA) to the nodes in the network, while satisfying interference constraints. The commonly used interference constraint (which is admittedly based on a somewhat idealized channel model) requires that no nodes within two hops of each other may use the same slot.

The two-hop constraint is needed to prevent hidden terminal collisions. Most of the scheduling schemes that have been proposed and studied in the context of WSN have been TDMA techniques, so we shall largely restrict our attention to these, but FDMA/CDMA-based scheduling techniques involve essentially similar channel assignment problems. While traditional TDMA assignment techniques do not specifically target energy conservation, it is almost trivial to get energy savings with these techniques because of their predictable communication patterns – all that needs to be done is to ensure that each node is asleep whenever it is not scheduled to transmit or receive. It is well known that the problem of assigning a minimum number of channels to an arbitrary graph, while ensuring that the two-hop constraints are satisfied, is an NP-hard coloring problem [169]. However, it is possible to provide efficient solutions in practice using heuristic assignments. One approach to time slot assignment is to perform it in a centralized manner, requiring that a base station gather the full network topology, perform the assignment offline, then distribute it back to the network [31]. However, such solutions do not scale well with network size, particularly in the case of dynamic environments. Decentralized approaches are therefore called for.

1. Stationary MAC and startup (SMACS)

One decentralized approach is the stationary MAC and startup algorithm proposed in. In this algorithm, each node need only maintain local synchronization. During the starting phase, each node decides on a common communication slot with a neighboring node through handshaking on a common control channel. Each link also utilizes a unique randomly chosen frequency or CDMA frequency hopping code. It is assumed that there are sufficiently many frequencies/codes to ensure that there are no common frequency/time assignments within interference range, and hence there is no contention. The slot is then used periodically, once each cycle, for communication between the two nodes.

2. BFS/DFS-based scheduling

One approach to performing channel assignment is the use of breadth-first (BFS) or depth-first (DFS) traversals of a data-gathering tree. In these protocols, every single node is given a unique slot, i.e. there is no spatial reuse of slots. It ensures that each node is given as many slots as the load that it needs to carry. The BFS- and DFS-based slot allocation schemes are illustrated in Figure 6.9. There are interesting tradeoffs with each approach. With BFS, each node gets contiguous time slots, which is advantageous if the energy costs to switch a radio

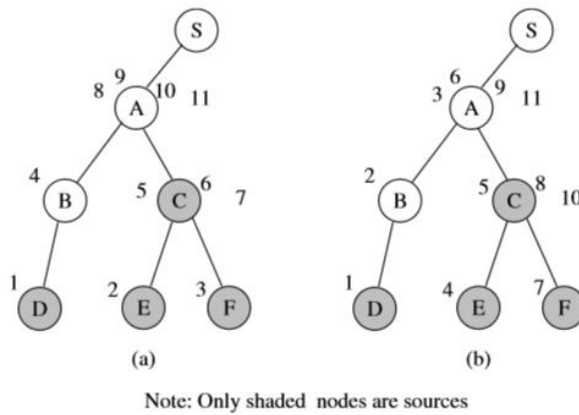


Figure 6.9 Time slot allocations for a data-gathering tree: (a) BFS, (b) DFS

from active to sleep and vice versa are large, since it minimizes such transitions at each node. With DFS, each node does not have contiguous slots, but the slots from each sensor source to the sink are contiguous, ensuring that intermediate node buffers are not filled up during data-gathering. This provides low latency to some extent; although the data must remain queued at the sensor node until its transmission slot, once the data is transmitted it is guaranteed to reach the sink in as many steps as the number of hops. BFS has also been used for more tightly “packed” channel assignment for a data-gathering tree in a scenario where interference constraints are taken into account to provide spatial reuse. In this slot-allocation scheme, each node performs slots assignment sequentially. At each node’s turn, it chooses time slots from the earliest possible slot number for its children. Local message exchanges ensure that the slot does not interfere with any already assigned to nodes within two hops of the child. The number of slots to be assigned to each node is pre-determined based on an earlier max–min fair bandwidth allocation phase. With these techniques, though, it should be kept in mind that they generally require global synchronization.

3. Reservation-based synchronized MAC (ReSync)

A significant concern with many of the TDMA schemes that provide guaranteed slots to all nodes is that they are not flexible in terms of allowing the traffic from each node to change over time. Reservation-based schemes such as ReSync provide greater flexibility. In ReSync, each node in the network maintains the notion of an epoch based on its local time alone, but it is assumed that each node’s epoch lasts the same duration (or can be synchronized with nearby neighbors accordingly). Each node picks a regular time each epoch based on its local clock to send a short intent message. It is assumed that this selection can be done without significant collisions because the intent message duration is very short. By listening for sufficiently long, each node must further learn when its neighbors send intent messages so that it can wake-up in time to listen to them. When a node wishes to transmit to another node, it indicates in the intent message when it plans to transmit the data (this data transmission time is chosen randomly and indicated as an increment rather than in absolute terms). The intended recipient will then wake-up at the corresponding time (which it can calculate based on its own local clock) in order to receive the message. ReSync does not incorporate an RTS/CTS mechanism to prevent message

collisions due to the hidden node problem; however, since the data transmissions are scheduled randomly, any collisions are not persistent.

4. Traffic-adaptive medium access (TRAMA)

The traffic-adaptive medium access (TRAMA) protocol is a distributed TDMA technique that allows for flexible and dynamic scheduling of time slots. Time epochs are divided into a set of short signaling slots, followed by a set of longer transmission slots. TRAMA consists of three key parts:

1. Neighbor protocol (NP): Nodes exchange one-hop neighboring information during the random access signaling slots. The signaling slots are sufficiently long to ensure that this information is reliably transmitted and all nodes have consistent two-hop neighborhood information.

2. Schedule exchange protocol (SEP): Each node publishes its schedule during its last winning slot in each epoch. The schedule uses bitmaps to efficiently denote the intended recipients (unicast or broadcast) for each of its future packets in the next epoch. If the number of packets a node wishes to send is less than the number of winning slots in an epoch, it gives up the remaining slots (which are packed at the end together between the last used winning slot, referred to as the changeover slot, and the final schedule propagation slot). The changeover slot is used for synchronization; therefore all neighbors are required to wake-up to listen to it. The node sleeps at all times when it is not required to transmit or receive.

3. Adaptive election algorithm (AEA): This algorithm uses a hash function based on node IDs and time to ensure that there is a unique ordering of node priorities among all nodes within any two-hop region at each time. In the simplest version of this algorithm, a node transmits if and only if it has the highest priority among all nodes within its two-hop neighborhood. However, to prevent the waste of a slot, if there is no data to send, TRAMA contains a sophisticated extension that allows for slot reuse. The slot reuse capability of TRAMA involves the use of additional information from a given node u 's point of view. The absolute winner for u is the node within two hops of u that has the highest priority; an alternate winner for a node u is a node within u 's one-hop neighborhood that has the highest priority; the possible transmitter set is the set of all one-hop neighbors of u who have highest priority among all their two-hop neighbors that are neighbors of u . A need transmitter is the node with the highest priority among u and the possible transmitter set, such that they need additional transmission slots. From a node's perspective, the absolute winner is the assumed transmitter unless the alternate winner is hidden from the absolute winner and is in the possible transmitter set, in which case the assumed transmitter is the alternate winner. Whenever the assumed transmitter gives up, the need transmitter is the true transmitter. Nodes that are not in the schedule listed by the transmitter can shift to sleep mode to save energy, while the relevant transmitter and receiver must stay awake to complete the pertinent communication. It is proved by Rajendran et al. that TRAMA is a correct protocol in that it avoids packet losses due to collisions or due to transmission to sleeping nodes.

Sleep-based topology control

The core functionality of a wireless sensor network depends on having a network topology that satisfies two important objectives: connectivity and coverage. Connectivity is essential to be able to move movement data between different points within the network, while coverage of the environment ensures that critical phenomena and events can be detected and monitored. The problem of constructing an initial topology is of course the problem of node deployment. Beyond this, there are many scenarios in which topology control is desired. In its broadest sense, topology control can be defined as the process of configuring or reconfiguring a network's topology through tunable parameters after deployment. There are three major tunable parameters for topology control:

1. **Node mobility:** In WSNs consisting of mobile nodes, such as robotic sensor networks, both coverage and connectivity can be adapted by moving the nodes accordingly.
2. **Transmission power control:** In WSNs with static nodes, if the deployment density is already sufficient to guarantee the required level of coverage, the connectivity properties of the network can be adjusted by tuning the transmission power of the constituent nodes.
3. **Sleep scheduling:** Finally, consider large-scale static WSNs deployed at a high density, i.e. over deployed. In this case, the appropriate topology control mechanism that provides energy efficiency and extends network lifetime is to turn off nodes that are redundant.

The remaining nodes form the active network over which sensing, computation, and communication take place. The challenge is to periodically determine if any active nodes have malfunctioned or if they have suffered energy depletion, so that additional sleeping nodes can be activated if needed. Sleep-based topology control mechanisms are the subject of this chapter.

Certainly, in applications(e.g.in industrial process monitoring)where a limited number of expensive nodes must be deployed in specific locations and must all be active at all times, it is unlikely that redundant nodes will be deployed. Thus sleep-based topology control schemes are not always relevant. However, in other contexts, where (a) the nodes are relatively inexpensive (so that the cost of redundancy is not unreasonably high), (b) the deployed location is remote (so that adding additional nodes at a later time is either undesirable or infeasible), and (c) the precise positioning of each active sensors is not essential, over deploying the network may provide some significant advantages: 1. Longer lifetime: The first is that an over-deployed network, if carefully managed through the topology control mechanisms described in this chapter, can have a longer network lifetime. This is obtained by keeping redundant nodes asleep until they are needed to replace the nodes that have failed or depleted their energy. 2. Robustness: While one could argue that longer lifetimes could also be obtained for a lower cost merely by adding longer-lifetime batteries, deploying redundant nodes also provides robustness to device failures for reasons other than energy depletion. 3. Tunable connectivity/coverage: Another significant advantage of over deploying a network is that it enables the possibility of adaptive topology control techniques that provide tunable levels of connectivity and coverage.

To save energy and maximize lifetime for an over-deployed network, it is essential to ensure that only the minimal set of nodes needed are awake at any time. The goal of sleep-based topology control is to determine which nodes should be awake at a given time to serve both

connectivity and coverage needs. Because of the unattended nature of these networks, it is essential that the topology control process be self-configuring and adaptive. In most cases, distributed, localized techniques are essential, to minimize overhead and enable quick reaction to changes. The basic approach used in many distributed sleep-based topology control techniques for WSN is the following. A set of sleep-related states is defined for each node, and the nodes then transition between these states depending on explicit messages with their neighbors, overheard messages/beacons, or implicit timers. Consider the following simple example as an illustration. Each node can be in one of three states: sleep, test, and awake. By default a node switches periodically, using a timer, between sleep and test states. During the test state, a local eligibility rule is applied (e.g., “are there less than K awake neighbors?”, “is there data intended for me?”) to determine whether the node should wake-up. If the rule is satisfied, then the node transitions to the wake state. When the node is an active state, it may (a) remain there forever, if the goal is to ensure that the active backbone of the network remains up at all times, or (b) return back to sleep after some timer expires; for instance, if the eligibility criterion is no longer satisfied.

Constructing topologies for connectivity

1. Basic/adaptive fidelity energy conserving algorithms (BECA/AFECA)

The basic energy-conserving algorithm (BECA) has three states: sleep, listen, and active. It is illustrated in Figure 7.1(a). The eligibility criterion used to transition from listen to active state is the presence of routing traffic in which the node should participate. Different timers T_s and T_l are used to go from sleep to listen state and back. The node will also transition to sleep from active state if there is no relevant traffic for T_a units. The node can also transition directly from sleep to active state if it has outgoing data to send.

The closely related adaptive fidelity energy-conserving algorithm (AFECA) extends BECA by adapting the sleep time depending on the neighborhood density for increased sleep efficiency. Specifically, the sleep time is chosen as the basic sleep time multiplied by a random number between 1 and N , where N is the number of neighbors estimated in the listening period.

2. Geographic adaptive fidelity (GAF)

The geographic adaptive fidelity (GAF) technique, illustrated in Figure 7.1(b) uses sleep, discovery, and active states. The node moves from the

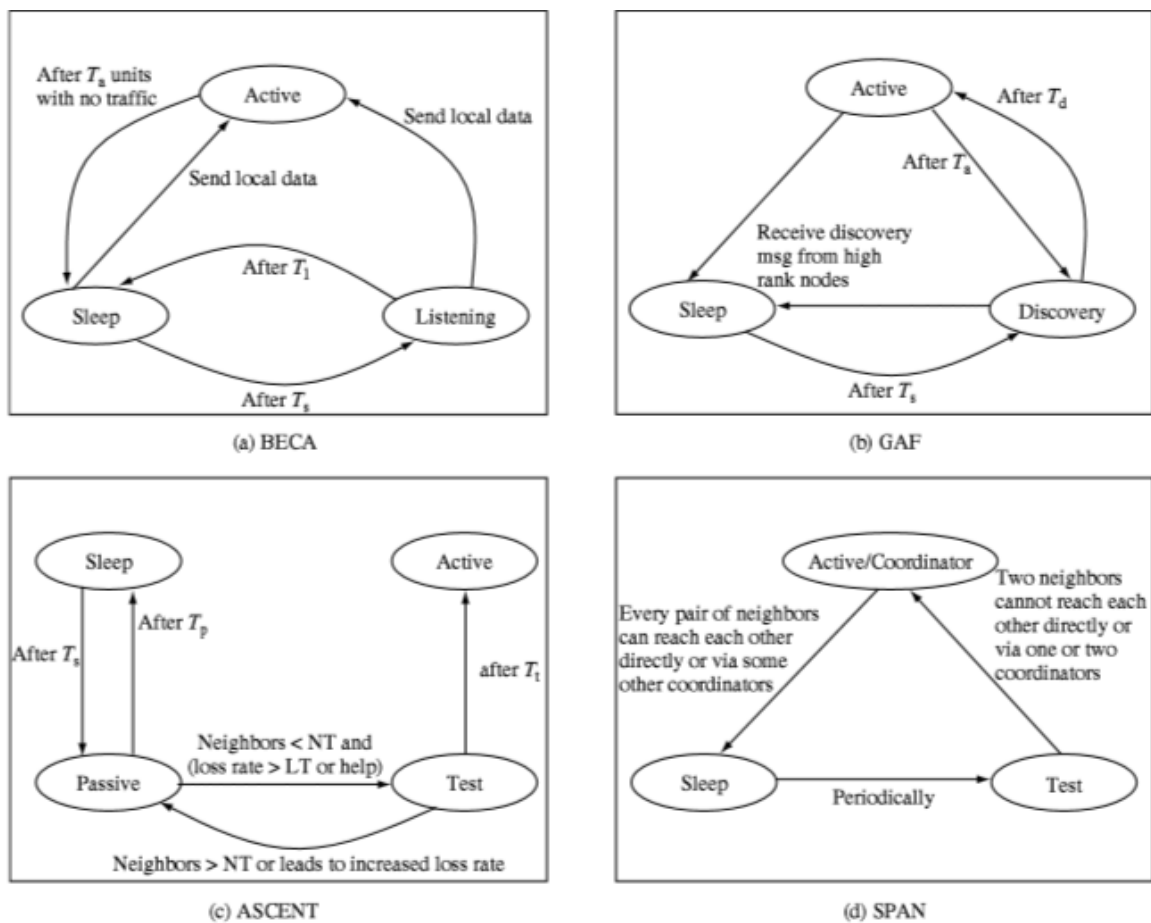


Figure 7.1 State transition diagrams for various sleep-based topology control techniques aimed at connectivity alone

discovery state to the active state and back with timers T_d and T_a respectively; and from the discovery state or the active state to the sleep state if it detects any other higher-priority nodes active within its geographical virtual grid (explained below) through the reception of a discovery message within its neighborhood. It moves from the sleep state to the discovery state after a time T_s . The nodes in discovery/active state transmit discovery messages containing their own ID, the grid ID, and estimated residual lifetime. Nodes with longer residual lifetimes have greater priority. The goal of GAF is therefore to ensure that a single node with the highest remaining lifetime is awake in each virtual grid. The virtual square grids are created to ensure that all nodes in one grid can communicate with all others in the adjacent grid.

3. Adaptive self-configuring sensor network topology control (ASCENT)

ASCENT is intended to adapt to highly dynamic environments; nodes wakeup to assist in routing depending on the number of active neighbors and the measured data loss rates in their vicinity. In ASCENT, illustrated in Figure 7.1(c), nodes are in one of four states: sleep, passive, test, and active. A node in a test state goes to the active state after a timer T_t unless it finds that the number of active neighbors is greater than the threshold NT or that its participation would increase the loss rate further (e.g. due to congestion), in which case it transitions back to the passive state. From the passive state the node transitions to the sleep state after a timer T_p unless the number of active neighbors is below the desired threshold and either the loss rate is higher than a threshold LT or a help message is received, in which case it transitions to the test

state. Help messages are sent by a node that needs the assistance of neighboring nodes to help relay messages to a given destination that it currently is unable to because of high loss. Finally, a node in the sleep state transitions to the passive state after a timer T_s . A key point to note in the basic ASCENT scheme is that nodes do not transition back to sleep from the full active state – this is suitable for applications where nodes in the constructed topology are needed/expected to be active until energy depletion. Message losses are assumed to be detected based on missing packet sequence numbers. There are a number of tunable parameters in ASCENT that must be adjusted depending on the application needs. The average degree and distance between active nodes needed for connectivity (or even coverage) influence NT and LT. The timers T_t and T_p effect a tradeoff between decision quality and energy efficiency, while T_s provides a tradeoff between energy efficiency and latency of response to failures in the network.

4. Neighborhood coordinators (SPAN)

SPAN is a related randomized approach to topology control that aims to provide connectivity with redundancy. Only a subset of nodes called coordinator nodes are in active state at any time. Its state transitions are illustrated in Figure 7.1(d). Nodes that are not coordinators go to sleep, waking up periodically to go to the test state to send HELLO messages and check for eligibility to become coordinators. The coordinator eligibility rule is the following: if two neighbors cannot reach each other directly or via one or two coordinators, then the node in test state should become an active coordinator. The decision regarding the eligibility rule is made based on the content of HELLO packets sent by all nodes in which they announce their current coordinators and current neighbors. This algorithm provides for a connected backbone, but not with a minimal set of nodes. It provides sufficient redundancy to ensure that there are coordinators in every radio broadcast range in the network, to minimize congestion bottlenecks. A randomized prioritized backoff is used to ensure that multiple nodes do not elect to become coordinators simultaneously. Nodes that are likely to benefit a greater set of neighbors are more likely to win contention to become coordinators.

Nodes withdraw as coordinators (i.e. go back from active to sleep state) if they no longer satisfy the eligibility rule, or after some timer period, to ensure load balancing.

Constructing topologies for coverage

The above-described topology control techniques have focused on waking up the radios of sufficient nodes to maintain good network connectivity. We now turn to the problem of ensuring good coverage in addition to connectivity. Most of these approaches are similar in spirit to those discussed above, except they involve coverage-based eligibility rules for activation of nodes.

1. Probing environment and adaptive sleep (PEAS)

The probing environment and adaptive sleep (PEAS) technique [235] aims to provide topology control for highly dynamic environments. There are three states in PEAS, as shown in Figure 7.2(a): sleep, probe, and active. From the sleep state the node uses a randomized timer with exponential distribution of rate to transition to the probe state. Randomized wake-up times are used to spread the probes from nodes so as to minimize the likelihood that any portion of the

network is left without an active node for too long. The rate d is adapted depending on the environment to ensure that the sleeping rate in every part of the network is about the same desired value d , regardless of spatio-temporal inhomogeneities in node density. In the probing state, a node detects if there is any active node within a probing range R_p , by sending a PROBE message at the appropriate transmit power and listening for REPLY messages. If there are no responses, the node transitions to the active state and stays there until its energy is depleted. If there are REPLY messages, the node computes an updated d , as described below, and transitions to the sleep state. Active nodes measure the rate at which PROBES are heard from nodes in their neighborhood to estimate the current aggregate neighborhood probing rate $\hat{d}$. The measured probe rate $\hat{d}$ and the desired probe rate d are sent in the REPLY message in response to each PROBE message.

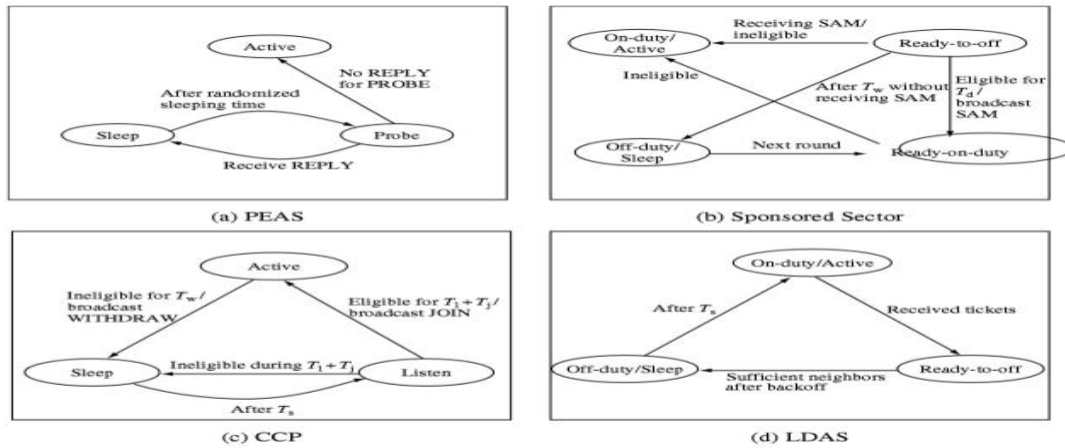


Figure 7.2 State transition diagrams for various sleep-based topology control techniques that address coverage

The choice of d depends on the application tolerance for delays, with the tradeoff being energy efficiency. Besides connectivity, the PEAS technique also provides tunable coverage. This is done through the eligibility rule in PEAS, where nodes try to ensure that there is an active neighbor within a configurable probing range R_p . R_p could be arbitrarily smaller than the communication range, and allows the density of active nodes to be varied. This does not necessarily guarantee, though, that the original coverage is preserved, which is the goal of the next scheme.

2. Sponsored sector

Tian and Georganas present an approach that aims to turn off redundant nodes while preserving the original coverage of the network. Each node checks to see if its coverage area is already covered by active neighbors before deciding whether to be on or off. Four states are used: off-duty, ready-on-duty, on-duty, ready-to-off, as seen in Figure 7.2(b). Nodes that are in the ready-on-duty state investigate whether they are eligible to turn off their sensor and radio, by examining their neighbors' coverage. If they are eligible, they first wait a random backoff time T_d , and broadcast a status advertisement message (SAM), before transitioning to the ready-to-off state. Any neighboring nodes with a longer backoff will not consider these nodes that issued a SAM before them in their neighbor coverage determination. This prevents multiple nodes from shutting off simultaneously. If nodes determine they are ineligible to turn off, they transition to the on-duty state. From the ready-to-off state, after a timer T_w , nodes transition to off-duty unless they receive a SAM from a neighbor and find that they are ineligible to turn off. If they are

ineligible to turn off, they transition to on-duty. The process is conducted via multiple sequential rounds. At the beginning of each round, each node sends a position advertisement message (PAM) to all neighbors within sensing range (by adjusting the transmission power accordingly), containing its sensing range as well as position. The coverage eligibility rule is determined by each node independently in the ready-on-duty state as follows. Based on the PAM messages, each node determines which angular sector of its own sensing range is covered by each of its neighbors (that have not already issued SAMs). If the union of all these sponsored sectors is equal to the node's own coverage area, it determines itself to be eligible to turn off. At the end of each round, eligible nodes turn off, while other nodes continue to sense.